

The Emergence of Geometry

from a Quantum Family Tree

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Abstract

We extend the entropy-rate theorem of the Quantum Family Tree (QFT) model from the binary qubit case ($p = 2, d = 2$) to general p -ary trees with d -dimensional bonds. Four structural theorems are proved: (i) a fixed-point convergence theorem establishing $\rho_* = 1/d$ as the unique stationary state with L^2 contraction rate $\eta_2(p, d)$; (ii) the first-law identity $\delta\langle H_A \rangle / \delta S = 1$ exact at all scales; (iii) a geometric precondition for the Lewkowycz–Maldacena derivation, with corrections exponentially small at rate η_2 ; and (iv) the resolution of the annealed Rényi-2 slope as $8/27$, with finite-depth corrections decaying at rate $\eta_4/\eta_2 = 13/28$. The Weingarten purity recursion is derived exactly for all (p, d) : $\alpha = \eta_2(p, d) = d^{p-1}(d^2 - 1)/(d^{2p} - 1)$, $\beta = d(d^{2(p-1)} - 1)/(d^{2p} - 1)$, with a remarkable (α, β) swap between complementary cuts. The Klein–Gordon graph-Laplacian mass formula is falsified at $p \geq 3$. Numerical measurements on the ternary ($p = 3$) tree support the universality conjecture $w_2(p, d) = (1 - \eta_2)/2$, with the factor $1/2$ independent of p . The universality is reduced to a single scalar identity—the one-boundary quadratic susceptibility—through a Reduction Theorem that expresses the entropy rate as $h_{VN} = 2\chi V_*(1 - \eta_2)$, and verified numerically to within 5%. The scalar identity remains conjectural at both the binary level (Lemma D of Paper 1) and the general (p, d) level; closing Lemma D would fix the binary susceptibility at $\chi_{2,2} = \frac{3}{2}\log_2(5/2)$.

Introduction

In Paper 1, *The Emergence of Distance from a Quantum Family Tree*, the author derived a structural reduction for the von Neumann entropy coefficient $c_{VN} = m^2 \cdot c_{RT}$ for the binary Haar tree ($p = 2, d = 2$), with $m^2 = w_2 = (1 - \eta_2)/2 = 3/10$ (the Lemma D conjecture, numerically supported at 10.34σ) and $\eta_2 = 2/5$. In that model, Haar-random $U(4)$ gates are placed on the edges of a binary Bruhat–Tits tree of depth D , producing a random boundary state on 2^D qubits. The entanglement entropy of a contiguous boundary interval of 2^k leaves grows linearly as $S_{VN} \approx (c_{VN}/3) \cdot k$, with the coefficient $c_{VN} = (9/10)\log_2(5/2) \approx 1.19$ bits, conditional on the sector-weight identity (Lemma D). The proof chain—Weingarten eigenvalue $\rightarrow$ mode budget $\rightarrow$ annihilation $\rightarrow$ exchange symmetry $\rightarrow$ cut averaging—reduces the entropy-rate theorem to a single scalar quantity: the sector weight w_2 .

This paper addresses the natural generalization: does the conjectured formula $w_2 = (1 - \eta_2)/2$ extend to p -ary trees with d -dimensional bonds? We prove exact Weingarten formulas for the purity recursion at all (p, d) , establish four structural theorems, and present numerical evidence supporting the universality conjecture:

$$w_2(p, d) = \frac{1 - \eta_2(p, d)}{2}$$

where $\eta_2(p, d) = d^{p-1}(d^2 - 1)/(d^{2p} - 1)$ is the Haar-averaged purity eigenvalue for $U(d^p)$ gates.

The factor $1/2$ is independent of p . This is established through a boundary-mode decomposition: the entropy increment at each scale is the sum of two localized boundary responses (one per interval endpoint) plus an exponentially screened bulk remainder. The symmetry of the two boundary channels forces the $1/2$ factor, independent of the tree branching structure. This reduces the conjecture, for all (p,d) , to the single scalar identity $\chi_{\{p,d\}} \cdot V^* = (1/4) \log_2(1/\eta_2)$.

Outline. Section 2 derives the general Weingarten purity formulas. Section 3 proves the fixed-point theorem. Section 4 establishes the first law at all scales. Section 5 proves the geometric precondition. Section 6 develops the boundary-mode decomposition and presents the universal sector weight conjecture. Section 7 resolves the Rényi-2 slope. Section 8 discusses the Padé continuation. Section 9 is the discussion.

General Weingarten Purity Recursion

The purity eigenvalue $\eta_2(p, d)$

Theorem 1 (General purity recursion). *For $V \sim \text{Haar}(U(N))$ with $N = d \cdot d_E \cdot d_E = d^{p-1}$,*

the single-child channel $\Phi(\rho) = \text{Tr}_E[V(\rho \otimes |0\rangle\langle 0|)V^\dagger]$ satisfies

$$E[\text{Tr}(\Phi(\rho)^2)] = \alpha \text{Tr}(\rho^2) + \beta$$

$$\text{with } \alpha(p, d) = \eta_2(p, d) = \frac{d^{p-1}(d^2-1)}{d^{2p}-1}, \quad \beta(p, d) = \frac{d(d^{2(p-1)}-1)}{d^{2p}-1}.$$

Proof. Direct application of the Weingarten second-moment formula for $U(N)$. The output purity involves four terms from the $(\pi, \sigma) \in S_2 \times S_2$ sum. With $Wg(id, N) = 1/(N^2 - 1)$ and $Wg((12), N) = -1/(N(N^2 - 1))$, collecting output and input factors yields the stated α and β . Both reduce to $d/(d^2 + 1)$ at $p = 2$, confirming $\alpha = \beta = \eta_2$ for the binary tree. $\square$

Key values

(p, d)	$\eta_2 = \alpha$	β	$p_{stat} = \beta/(1 - 1 - \eta_2)$	
(2, 2)	2/5	2/5	2/3	3/5
(3, 2)	4/21	10/21	10/17	17/21
(4, 2)	8/85	42/85	6/11	77/85
(2, 3)	3/10	3/10	3/7	7/10

Self-duality at $p = 2$. When $d_E = d$ (binary tree), $\alpha = \beta = d/(d^2 + 1)$. For $p > 2$, $\beta > \alpha$: the environment contribution dominates.

The (α, β) swap

Proposition 2 (Cut complementarity). *For a $(j | p - j)$ cut with system dimension $d_s = d^j$*

$$\text{and environment dimension } d_E = d^{p-j}: \alpha_{j|p-j} = \frac{d_E(d_s^2 - 1)}{N^2 - 1}, \quad \beta_{j|p-j} = \frac{d_s(d_E^2 - 1)}{N^2 - 1}.$$

The $(1|2)$ and $(2|1)$ cuts at $p = 3$ give $(\alpha, \beta) = (4/21, 10/21)$ and $(10/21, 4/21)$ respectively: the coefficients swap under cut complementarity.

Falsification of the graph-Laplacian mass formula

A natural conjecture, motivated by the Gubser et al. p -adic AdS/CFT correspondence, is that the effective mass can be read off from the graph Laplacian of the tree:

$m^2(p, d) = (d^2 - 1)(d^2 + 1 - 2p)/(2(d^2 + 1))$. This “KG formula” (named for the Klein–Gordon equation on the Bruhat–Tits tree) gives $m^2 = 3/10$ at $(p = 2, d = 2)$, matching the target value. However, at $(p = 3, d = 2)$ it gives $m^2 = -3/10$. A negative mass-squared is unphysical, and the formula is falsified for all $p \geq 3$ at $d = 2$. The failure indicates that the relationship between tree geometry and entanglement structure is subtler than the graph-Laplacian eigenvalue alone captures.

Fixed-Point Theorem

Theorem 3 (Fixed-point convergence). *Let $V \sim \text{Haar}(U(d^2))$,*

$\Phi_V(\rho) = \text{Tr}_2[V(\rho \otimes |0\rangle\langle 0|)V^\dagger]$, *and $T_V \in M_{d^2-1}(R)$ the Bloch map. Then:*

1. $\Phi^-(\rho) = I/d$ *for all ρ (completely depolarizing).*
2. $E[T_V^\dagger T_V] = \eta_2 I_{d^2-1}$, *giving $E[\|T_n \cdots T_1 r_0\|^2] = \eta_2^n \|r_0\|^2$.*
3. $\|T_n \cdots T_1\| \rightarrow 0$ *almost surely (Borel–Cantelli).*
4. *Unique bidirectional fixed point $\rho_*^\downarrow = \rho_*^\uparrow = I/d$.*

Proof. (a) Left $U(d)$ covariance $\Rightarrow \Phi^-(\rho) = c \cdot I$ by Schur's lemma; trace preservation gives $c = 1/d$. (b) Same covariance forces $E[T^\dagger T] \propto I$; trace gives the proportionality constant η_2 ; iterate by independence. (c) Markov + Borel–Cantelli. (d) SWAP invariance of the Haar measure exchanges the output qudits, implying $\rho_*^\downarrow = \rho_*^\uparrow$. $\square$

Contraction constant. $\eta_2(p = 2, d = 2) = 2/5$. Quenched Lyapunov exponent

$\lambda_1 \approx -0.572$ nats/step (measured, 2000 trials at $n = 100$).

First Law at All Scales

Theorem 4 (Universal first law). *For any density matrix ρ_A and any traceless perturbation*

$$\delta\rho_A: \delta\langle H_A \rangle = \delta S_{VN}$$

identically, at all k , all positions, for any gate ensemble.

Proof. $\delta S_{VN} = -\text{Tr}[\delta\rho_A(\log\rho_A + I)] = -\text{Tr}[\delta\rho_A \log\rho_A] = \delta\langle H_A \rangle$, since $\text{Tr}[\delta\rho_A] = 0$. $\square$

Remark 5. The identity carries no geometric content beyond the definition of the modular Hamiltonian. The interesting question is whether the geometric precondition (Section 5) allows this algebraic identity to be promoted to a gravitational derivation.

Geometric Precondition

Theorem 6 (Exponential convergence). *For the p -ary Haar tree of depth D , the entropy*

$$S_{VN}(A) \text{ for an interval of } p^k \text{ leaves satisfies: } E\left[\left|S_{VN}^{(D)}(A) - S_{VN}^{(\infty)}(A)\right|^2\right] \leq C(k)^2 \cdot \eta_2^{D-k}.$$

Proof. By Theorem 3, the carry at depth $D - k$ has converged to within $O(\eta_2^{D-k})$ of the stationary distribution. The Choi recursion maps carry perturbations to boundary entropy perturbations with a k -dependent Lipschitz constant. $\square$

Numerical confirmation. For the binary tree: $S_{VN}(k = 1)$ converges to $S_{stat} = 0.744$ bits by $D - k \approx 5$, with deviations matching the affine recursion $p_d = 2/3 + (1/3)\eta_2^d$.

Consequence. The Lewkowycz–Maldacena gravitational entropy derivation applies on the infinite Bruhat–Tits tree ($D = \infty$) with exact geometric precondition. On finite trees, corrections are below 1% for $D - k \geq 5$.

Boundary-Mode Decomposition and the Universal Sector Weight

Structural claim

The entropy increment per scale cannot be understood as an uncorrelated average over children corrected by mutual information. The correct abstraction is that entropy production is carried by two boundary modes, one for each endpoint of the interval. This is the invariant statement compatible with: (i) the isotropic transfer contraction $E[T^\dagger T] = \eta_2 I$ (Theorem 3); (ii) the interval having exactly two boundary rays in the tree geometry; and (iii) the binary proof already yielding $w_2 = (1 - \eta_2)/2$.

Boundary decomposition of the entropy increment

Proposition 7 (Boundary decomposition). *Let A_k be an interval of p^k leaves in the infinite p -ary Haar tree, and let $\Delta_k := S(A_{k+1}) - S(A_k)$ denote the entropy increment per scale. Then*

$$\text{there exist random variables } J_L^{(k)}, J_R^{(k)}, \text{ and } \varepsilon_k \text{ such that } \Delta_k = J_L^{(k)} + J_R^{(k)} + \varepsilon_k,$$

where $J_L^{(k)}$ depends only on gate data near the left boundary ray, $J_R^{(k)}$ on the right boundary ray, and ε_k is a bulk remainder from correlations among children at the two straddle nodes.

Let $\rho_{L,k}^{in}$ and $\rho_{R,k}^{in}$ be the incoming carry states on the boundary edges entering the left and right straddle nodes, and let $F_{p,d}(\rho)$ be the Haar-averaged entropy response functional of one straddle node. Define

$$J_L^{(k)} := F_{p,d}(\rho_{L,k}^{in}) - F_{p,d}(\rho_*), \quad J_R^{(k)} := F_{p,d}(\rho_{R,k}^{in}) - F_{p,d}(\rho_*),$$

where $\rho_* = I/d$. The remainder ε_k captures the failure of exact additivity when both straddles are present simultaneously.

Symmetry of the boundary channels

Proposition 8 (Boundary symmetry). *By reflection symmetry of the interval and permutation symmetry of the p children under Haar averaging: $E[J_L^{(k)}] = E[J_R^{(k)}]$.*

Proof. The Haar measure on $U(d^p)$ is invariant under child-permutation symmetry. The left and right boundary rays are exchanged by reflection of the interval. Under this exchange, the gate and carry distributions are identical. $\square$

Bulk screening

Proposition 9 (Screening of interior modes). *Let X_k denote the traceless perturbation carried into the scale- k straddle region from the interior. Then $E \| X_{k+1} \|^2 = \eta_2 E \| X_k \|^2$.*

Any entropy contribution from interior traceless data is exponentially screened and does not contribute to the asymptotic coefficient except through the stationary boundary injection.

Proof. Direct consequence of Theorem 3(b): $E[T^\dagger T] = \eta_2 I$ forces geometric contraction of the traceless component at rate η_2 . $\square$

Quadratic boundary response

Since I/d maximizes entropy, the first variation of the entropy functional vanishes identically on the traceless subspace: $\delta S|_{\rho_*} = 0$. The leading boundary response is therefore quadratic.

Lemma 10 (Isotropic quadratic response). *The one-boundary entropy response functional*

$$\text{satisfies } F_{p,d}(\rho_* + \delta\rho) = F_{p,d}(\rho_*) - \chi_{p,d} \text{Tr}(\delta\rho^2) + O(\|\delta\rho\|^3),$$

where $\chi_{p,d} > 0$ is a scalar susceptibility. The quadratic form is isotropic by Haar symmetry.

$$\text{The susceptibility decomposes as } \chi_{p,d} = \frac{d}{2\ln 2} \kappa_{p,d},$$

where $d/(2\ln 2)$ is the universal entropy Hessian at I/d and $\kappa_{p,d}$ is the isotropic

second-moment coefficient of the one-straddle boundary map $B_{p,d}$:

$$E\left[Tr\left(B_{p,d}(\delta\rho)^2\right)\right] = \kappa_{p,d} Tr(\delta\rho^2).$$

Reduction Theorem. Using the fixed-point theorem, Haar isotropy, and the two-boundary decomposition, the asymptotic entropy slope takes the structural form

$$h_{VN} = 2\chi_{p,d} V_*(1 - \eta_2).$$

The factor of 2 arises precisely from the two boundary endpoints of the interval: by Proposition 6.2, $E[J_L] = E[J_R]$, so the total rate is twice the single-boundary contribution.

Here $V_* = E\left[Tr\left((\rho_{carry} - I/d)^2\right)\right] = p_{stat} - 1/d$ is the stationary traceless variance of the boundary carry.

The universality conjecture $w_2 = (1 - \eta_2)/2$ is equivalent to the single scalar identity

$$\chi_{p,d} V_* = \frac{1}{4} \log_2(1/\eta_2).$$

The universal sector weight

Conjecture 11 (Boundary equipartition). *The one-boundary susceptibility satisfies:*

$$\chi_{p,d} V_* = \frac{1}{4} \log_2(1/\eta_2).$$

Equivalently, by the Reduction Theorem: $w_2(p, d) = (1 - \eta_2(p, d))/2$.

The entropy scale $\log_2(1/\eta_2)$ appears because η_2 is the unique nontrivial eigenvalue controlling decay of traceless modes on the carry, and $\log_2(1/\eta_2)$ is the associated information decay rate per level.

Conditional binary susceptibility

For $(p, d) = (2, 2)$, the structural ingredients are exact: $\eta_2 = 2/5$, $p_{stat} = 2/3$, and $V_* = 1/6$.

The Reduction Theorem gives $h_{VN} = \chi/5$. The binary entropy slope

$h_{VN} = (3/10)\log_2(5/2)$ is derived conditionally in *The Emergence of Distance* (Paper 1,

Theorem 8). Inverting:

$$\chi_{2,2} = \frac{3}{2}\log_2(5/2) \approx 1.9829.$$

Equivalently, $\kappa_{2,2} = \frac{3}{2}\ln(5/2) \approx 1.3745$, and $\chi_{2,2}V_* = \frac{1}{4}\log_2(5/2)$, which verifies the

scalar identity of Conjecture 6.4 at $(p,d) = (2,2)$ conditional on Lemma D. Closing Lemma D would make $\chi_{2,2}$ an unconditionally derived quantity and fix the template for the general (p,d) case.

For general (p, d) , the conjecture gives:

$$\chi_{p,d} = \frac{\frac{1}{4}\log_2(1/\eta_2)}{p_{stat}^{-1/d}}.$$

At $(3, 2)$: $\chi_{3,2} = \frac{17}{6}\log_2(21/4) \approx 6.778$. The susceptibility grows with p because the stationary carry moves closer to I/d (V_* shrinks), requiring steeper curvature of the entropy landscape to maintain the same entropy production rate.

Numerical confirmation

Remainder decay. Using the *Emergence of Distance* (Paper 1) MPDO model data, the remainder $\varepsilon_k = \Delta_k - h_{VN}$ satisfies:

k	Δ_k	ε_k	$ \varepsilon_k /h_{VN}$
1	0.3931	− 0.0035	0.88%
2	0.3991	+ 0.0025	0.64%

The boundary decomposition is better than 99% accurate from $k = 1$ onward. The remainder is bounded by the subleading eigenvalue ratio: $\varepsilon_k = O\left((\eta_4/\eta_2)^k\right)$ with $\eta_4/\eta_2 = 13/28$, consistent with the weighted least-squares (WLS) fit residual structure from Paper 1.

Carry-chain measurement at $p = 3$. A carry-chain MPDO simulation on ternary trees ($U(8)$ Haar-random gates, bond dimension $\chi = 2$, ancilla dimension $d_E = 4$) with tree depths $D = 2-7$, maximum total qubits capped at 8, and 800 independent gate realizations per depth gives $w_2(p = 3) = 0.383 \pm 0.003$. Position averaging is uniform over all valid interval starting positions modulo the boundary size. The result is within 5% of the target $(1 - \eta_2)/2 = 17/42 = 0.405$. The systematic shortfall is attributed to state truncation. The competing candidates $(1 - \eta_2)(p - 1)/p = 0.540$ and $3/10 = 0.300$ are excluded at 29% and 28% respectively.

Two-chain calibration. Single-chain slope / $h_{VN} = 0.508$ (binary) and 0.479 (ternary), both consistent with the $1/2$ prediction from Proposition 6.2.

Relation to the cut-averaging argument

The naive cut-averaging argument gives a per-level fraction of $(p - 1)/p$, which equals $1/2$ only at $p = 2$. The boundary-mode framework explains the discrepancy: children at each straddle are correlated ($I(0:1) = 0.40$ bits at $p = 3$), and the mutual information between children placed in the same subsystem reduces the naive fraction back to $1/2$. In the boundary-mode language, this MI is absorbed into the remainder ϵ_k rather than serving as the proof mechanism.

Resolution of the Rényi-2 Slope

The position-averaged annealed Rényi-2 slope from the (b, c) recursion at $k = 10$ was reported as 0.2961 ± 0.0001 per doubling, with the closest rational candidates $8/27 \approx 0.2963$ and $37/125 = 0.2960$ not matching to 4 + digits.

Resolution. The asymptotic slope is exactly $8/27$. The finite- k correction comes from the subleading eigenvalue ratio $\eta_4/\eta_2 = 13/28$:

$$\Delta S_2(k) = \frac{8}{27} + A \cdot \left(\frac{13}{28}\right)^k,$$

with fitted amplitude $A \approx -0.42$. At $k = 10$: correction = -0.0002 , explaining the gap. By $k \geq 15$, the slope converges to $8/27$ at 5 + digits.

Padé Continuation and Exact h_3

The Padé $[1, 1]$ structure $h(q) = (a + bq)/(1 + 2q)$ with pole at $q = -1/2$ was established in *The Emergence of Distance* (Paper 1). Using exact $h_2 = 14/45$ and the target $h_1 = (3/10)\log_2(5/2)$ (assuming $w_2 = 3/10$), the Padé coefficients are determined:

$$b = \frac{14}{9} - \frac{9}{10}\log_2 \frac{5}{2}, \quad a = \frac{9}{10}\log_2 \frac{5}{2} - b.$$

This yields the exact quenched Rényi-3 rate:

$$h_3 = \frac{4}{9} - \frac{9}{70}\log_2 \frac{5}{2} = 0.27448 \dots$$

The value is **transcendental**, containing $\log_2(5/2)$, confirming that h_q for $q \geq 3$ cannot be a simple rational. The digamma approximation $h_3 = 11/40 = 0.275$ differs by 0.19%.

The h_3 formula is a **falsifiable prediction** of the hypothesis $w_2 = (1 - \eta_2)/2$. The full Padé curve gives predictions at all q : $h(4) = 0.25413$ (vs. digamma $413/1620 = 0.25494$, gap 0.08%).

Discussion

The boundary-mode framework

The central structural result of this paper is the boundary decomposition $\Delta_k = J_L + J_R + \varepsilon_k$.

The entropy increment per scale is the sum of two localized boundary responses plus an

exponentially screened remainder. The factor $1/2$ in $w_2 = (1 - \eta_2)/2$ is forced by the symmetry $E[J_L] = E[J_R]$, which holds because the interval has exactly two boundary rays exchanged by reflection.

The remaining analytical step is extending the one-boundary susceptibility from the conditionally derived binary value $\chi_{2,2} = \frac{3}{2} \log_2(5/2)$ to general (p, d) . The decomposition $\chi = (d/2 \ln 2) \cdot \kappa$ separates this into a universal entropy Hessian and a model-specific straddle coefficient $\kappa_{p,d}$, which is computable from the Haar integral over $U(d^p)$.

The (α, β) swap

The cut-complementarity relation—that the purity recursion coefficients (α, β) swap between a j -child cut and its complement—is a new structural feature of the p -ary tree. At $p = 2$, the swap is trivial ($\alpha = \beta$). At $p \geq 3$, it reflects a duality between the “system” and “environment” perspectives of the partial trace.

Open problems

Four problems remain:

1. *Sector-weight identity (Lemma D)*. The binary sector weight $w_2 = 3/10$, equivalently $\chi_{2,2} \cdot V^* = (1/4) \log_2(1/\eta_2)$, is the single open scalar identity across the trilogy. The structural reduction in Paper 1 isolates it; the universality conjecture of the present paper generalizes it. Numerical support is overwhelming: 10.34σ classical Monte Carlo ($N=2000$, $D=20-150$) and independent confirmation from a 156-qubit IBM Heron quantum run. Closing Lemma D analytically would upgrade the conditional

theorems in both papers to unconditional results. The problem is formulated as an exact 2D integral over the stationary carry density; see Paper 1 §8 for status.

2. *One-boundary susceptibility for general (p, d) .* The binary susceptibility $\chi_{2,2} = \frac{3}{2} \log_2(5/2)$ is derived conditionally (assuming Lemma D). Extending to general (p, d) requires computing the one-straddle boundary map coefficient $\kappa_{p,d}$ from the Haar integral over $U(d^p)$. This is a finite computation using Weingarten second and fourth moments.
3. *Verify h_3 from T_3 .* The Padé predicts $h_3 = 0.27448$. This should be verified by computing the quenched Rényi-3 rate from the T_3 eigenstructure $\{1, 2/5 (\times 3), 1/5\}$, or by high-statistics measurement at $k \geq 4$.
4. *K-P correction factor.* Define $\gamma(q) = h_q^{qu}/h_q^{ann}$. Then $\gamma(1) = 1$ and $\gamma(2) = 21/20$. The q -dependence of γ remains to be determined.
5. K. Donahue, *The Emergence of Distance from a Quantum Family Tree* (2026).